

Exercise 18.1

Q:1

Write out the characteristics
for each of the
following and find the
characteristics roots:

(a) $y_{t+2} - y_{t+1} + \frac{1}{2}y_t = 2$

$$a_1 = -1, a^2 = \frac{1}{2}, c = 2$$

characteristics roots:

$$b_1, b_2 = \frac{-a_1 \pm \sqrt{b^2 - 4a^2}}{2}$$

$$= \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1/2)}}{2}$$

$$= \frac{1 \pm \sqrt{1-2}}{2}$$

$$= \frac{1 \pm \sqrt{-1}}{2}$$

$$= \frac{1 \pm i}{2}$$

$$= \frac{1}{2} \pm \frac{1}{2}i \quad \underline{\underline{\text{Ans}}}$$

$$(b) \ yt+2-4yt+2+4yt=7$$

$$a_1 = -4, a_2 = 4, c = 7$$

characteristics roots

$$b_1, b_2 = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2}}{2}$$

$$= \frac{-(-4) \pm \sqrt{(-4)^2 - 4(4)}}{2}$$

$$= \frac{4 \pm \sqrt{16-16}}{2}$$

$$= \frac{4 \pm \sqrt{0}}{2}$$

$$= \frac{4 \pm 0}{2}$$

x_1

$$= 2 \pm 0$$

$$= \boxed{2} \quad \underline{\text{Ans}}$$

$$(c) \quad y_t + 2 + \frac{1}{2} y_{t+1} - \frac{1}{2} y_{t-1} = 5$$

$$a_1 = \frac{1}{2}, a^2 = -\frac{1}{2}, c = 5$$

characteristic roots

$$b_1, b_2 = \frac{-a_1 \pm \sqrt{a_1^2 - 4a^2}}{2}$$

$$= \frac{-\frac{1}{2} \pm \sqrt{(\frac{1}{2})^2 - 4(-\frac{1}{2})}}{2}$$

$$= \frac{-\frac{1}{2} \pm \sqrt{\frac{1}{4} + 2}}{2}$$

$$= \frac{-1/2 \pm \sqrt{9/4}}{2}$$

$$= \frac{-1/2 \pm 3/2}{2}$$

$$= \frac{-1 + \frac{3}{2}}{2}, \quad \frac{-1 - \frac{3}{2}}{2}$$

$$= \left[\frac{1}{2}, 2 \right], \quad \left[-1 \right]$$

$$= \left[\frac{1}{2}, 2 \right] \text{ ans}$$

$$(d) y_t + 2 - 2y_{t-1} + 3y_{t-2} = 4$$

$$a^2 = -2, a^2 = 3, c = 4$$

characteristics roots

$$b_1, b_2 = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2}}{2}$$

$$= \frac{-(-2) \pm \sqrt{(-2)^2 - 4(3)}}{2}$$

$$\begin{aligned}
 &= \frac{2 \pm \sqrt{4-12}}{2} \\
 &= \frac{2 \pm \sqrt{-8}}{2} \\
 &= \frac{2 \pm 2\sqrt{2}i}{2} \\
 &= \frac{2 \pm 2\sqrt{2(-1)}}{2} \\
 &= \frac{2 \pm 2\sqrt{-2}}{2} \\
 &= \frac{2 \pm 2\sqrt{-2}}{2} \\
 &= \boxed{1 \pm \sqrt{2}i} \quad \text{Ans}
 \end{aligned}$$

Q:4

Solve the following
difference equations:

(a) $y_t + 2 - 2y_{t-1} + 2y_{t-2} = 2$
($y_0 = 3; y_1 = 4$)

$$y_t + 2 - 2y_{t+1} + 2y_t$$

$$y_0 = 3$$

$$y_0 + 2 - 2y_{0+1} + 2y_0$$

$$= 3 + 2 - 2(3+1) + 2(3)$$

$$= 5 - 2(4) + 6$$

$$= 5 - 8 + 6$$

$$= 3$$

$$y_t + 2 - 2y_{t+1} + 2y_t$$

$$y_t = 4$$

$$= 4 + 2 - 2(4+1) + 2(4)$$

$$= 6 - 2(5) + 8$$

$$= 6 - 10 + 8$$

$$= 4$$

$$y_t + 2 - 2y_{t+1} + 2y_t = 1$$

$$a_1 = -2, a^2 = 2, c = 1$$

$$b_1, b_2 = h \pm v_i$$

$$h = \frac{-a'}{2}$$

$$h = \frac{-(-2)}{2} = \frac{2}{2}$$

$$h = [1]$$

$$V = \sqrt{4a'^2 - a''^2}$$

$$V = \frac{\sqrt{4(2)^2 - (-2)^2}}{2}$$

$$V = \frac{\sqrt{8-4}}{2}$$

$$V = \frac{\sqrt{4}}{2}$$

$$V = \frac{2}{2} = [1]$$

$$b_1, b_2 = 1, 1$$

$$R = \sqrt{h^2 + v^2}$$

$$= \sqrt{1^2 + 1^2}$$

$$R = \sqrt{1+1}$$

$$R = \boxed{\sqrt{2}}$$

$$\cos \theta = \frac{h}{R} = \frac{1}{\sqrt{2}}$$

$$\sin \theta = \frac{V}{R} = \frac{1}{\sqrt{2}}$$

See the table

$$C = \frac{\pi}{4}$$

Particular Solution:

$$y_p = \frac{1}{1+q_1+q_2}$$

$$y_p = \frac{1}{1+(-2)-2} = \frac{1}{-1}$$

$$y_p = \boxed{-1}$$

Complementary equations

$$y_c = e^{2t} (\cos \theta t + \sin \theta t) \\ = \sqrt{2} e^t \left(\cos \frac{\pi}{4} t + \sin \frac{\pi}{4} t \right)$$

General Solution:

$$y_t = y_c + y_p \\ = \sqrt{2} e^t \left(\cos \frac{\pi}{4} t + \sin \frac{\pi}{4} t \right) + 2$$

Ans

$$y_t + 2 - y_{t+1} + \frac{1}{4} y_{t+2} = 2 \\ (y_0 = 4; y_1 = 7)$$

$$y_0 + 2 - y_0 + 1 + \frac{1}{4} y_0 =$$

$$= 4 + 2 - 4 + 1 + \frac{1}{4} (4)$$

$$= \boxed{4}$$

$$y_1 + 2 - y_{1+1} + \frac{1}{4} y_1$$

$$= 7 + 2' - 7 + 1 + \frac{1}{4} (7)$$

$$= 8 - 8 + \frac{1}{4} (7)$$

$$= \boxed{7}$$

$$y_k + 2 - y_k + 1 + \frac{1}{4} y_k = 2$$

$$a_1 = -1, a^2 = \frac{1}{4}, c = 2$$

$$b_1, b_2 = h \pm vi$$

$$h = \frac{-a_1}{2} = \frac{-(-1)}{2} = \boxed{\frac{1}{2}}$$

$$V = \sqrt{4a^2 - a_1^2}$$

$$V = \sqrt{4\left(\frac{1}{4}\right) - (-1)^2}$$

$$V = \sqrt{\frac{0}{2}}$$

$$V = \boxed{0}$$

$$R = \sqrt{h^2 + V^2}$$

$$R = \sqrt{\left(\frac{11}{2}\right)^2 + 0}$$

$$R = \sqrt{1/4 + 0}$$

$$R = \sqrt{1/4}$$

$$R = 1/2$$

$$\cos \theta = \frac{h}{R} = \frac{1/2}{1/2} = 1$$

$$\sin \theta = \frac{V}{R} = \frac{0}{1/2} = 0$$

See the table

$$\phi = 2\pi \text{ or } 0$$

Particular Solution:

$$x_p = \frac{1}{1 + a_1 t + a_2}$$

$$y_p = \frac{1}{1 + (-1) + 1/4}$$

$$y_p = \frac{1}{4}$$

Complementary function:

$$y_c = e^{at} (\cos bt + \sin bt)$$

$$y_c = \left(\frac{1}{2}\right) e^{2t} (\cos 2\pi t + \sin 2\pi t)$$

General solution:

$$y_t = y_c + y_p$$

$$y_t = \left(\frac{1}{2}\right) e^{2t} (\cos 2\pi t + \sin 2\pi t) + \frac{1}{4}$$

Ans